L4

CLASS X - SCIENCE

LIFE PROCESSES

PRASHANT KIRAD

आरम्भ
BATCH
JOSH METER
KAISA HAI?
LOW
MODERATE
HIGH

TODAY'S TARGET

- Types of respiration
- Human Respiratory System
- Human Circulatory System
- Exchange of gases in plants

आरंभ BATCH

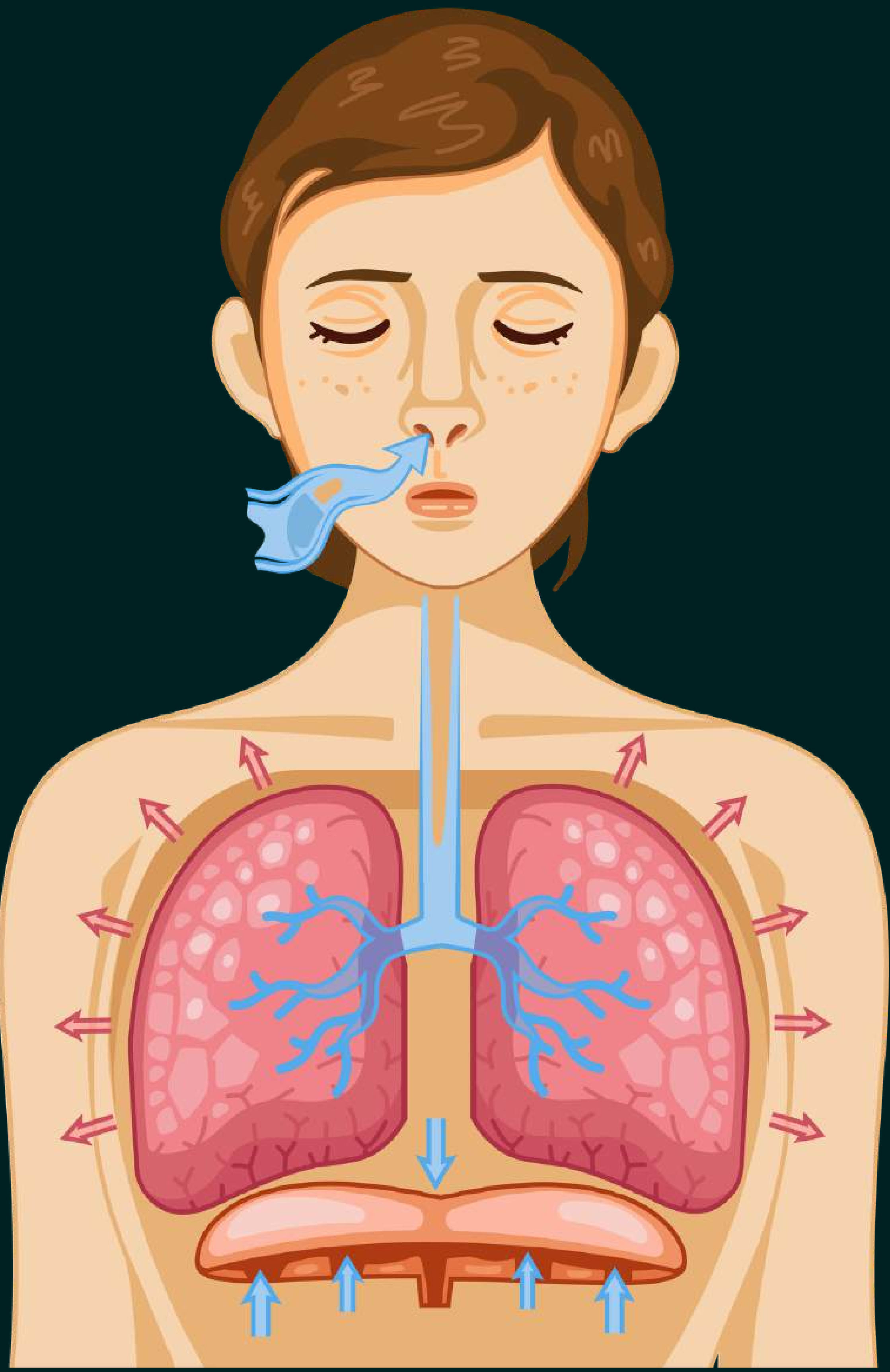
Respiration

Respiration is the process in which food is broken down inside the cells to release energy.

In this process, the energy stored in food is released and used by the body for different life processes.

1 Aerobic Respiration

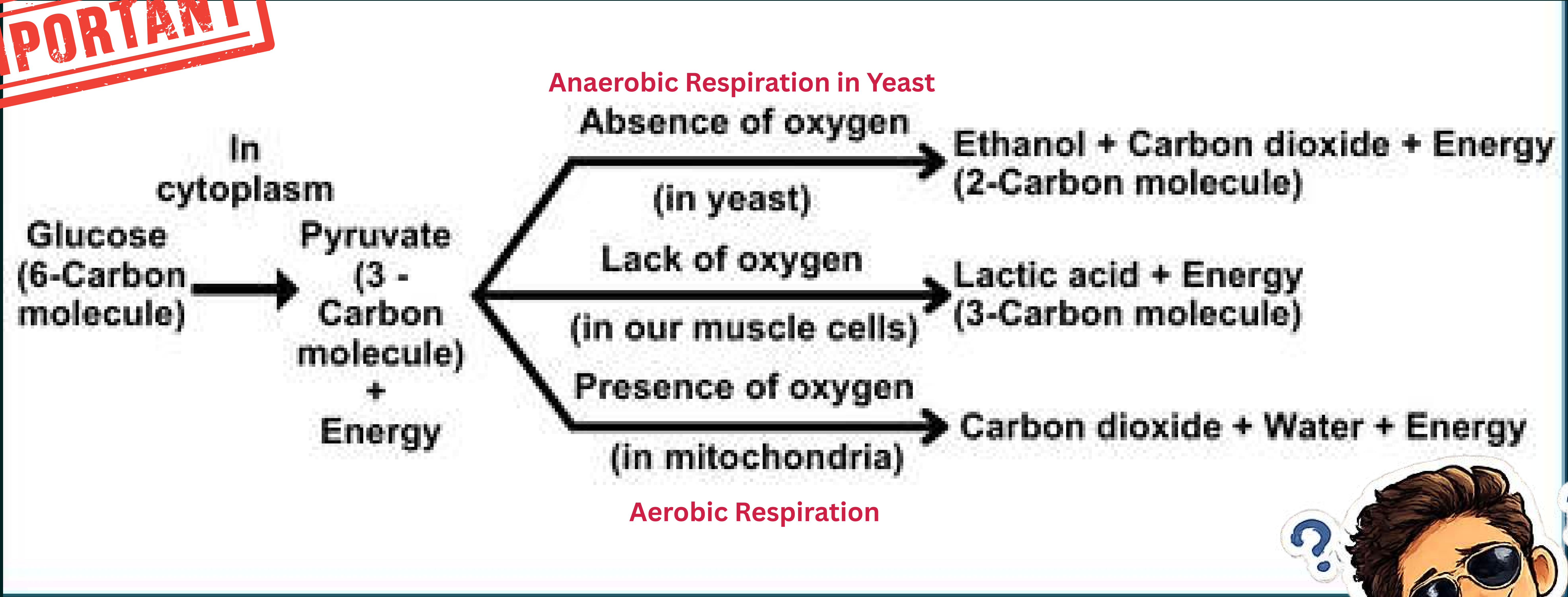
2 Anaerobic Respiration



Aerobic Respiration	Anaerobic Respiration
It takes place **in the presence of oxygen**.	It takes place **in the absence of oxygen**.
Final products are **carbon dioxide, water and energy**.	Final products may be **ethanol, carbon dioxide and energy** or **lactic acid and energy**.
It releases **more energy**.	It releases **less energy**.
It occurs mainly in the **mitochondria** after pyruvate formation.	It occurs in the **cytoplasm**.
It occurs in most plants and animals.	It occurs in organisms like **yeast** and also in our **muscle cells** during lack of oxygen.
Example: Normal respiration in humans.	Example: Fermentation in yeast and lactic acid formation in muscles.

Breakdown of Glucose

IMPORTANT



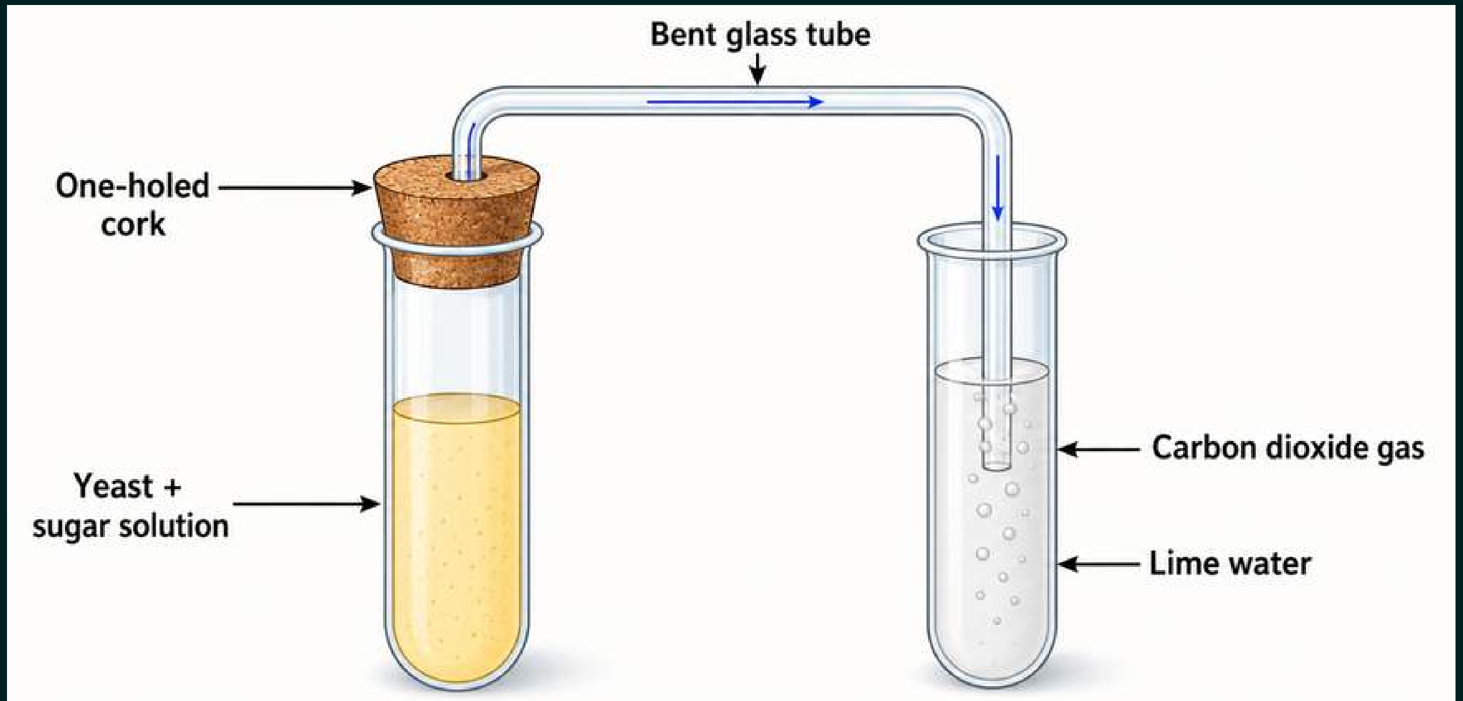
<u>**ATP:**</u> The energy released during respiration is used to make ATP.
ATP is used to fuel different activities in the cell.
When ATP breaks down, it releases a fixed amount of energy that helps in cellular activities.

Types of Anaerobic Respiration

Alcoholic Fermentation	Lactic Acid Formation
It occurs in the **absence of oxygen**.	It occurs when there is **lack of oxygen**.
Glucose is incompletely broken down to release energy.	Glucose is incompletely broken down to release energy.
It mainly occurs in **yeast**.	It occurs in **some bacteria** and **muscle cells**.
Products formed are **ethanol, carbon dioxide and energy**.	Products formed are **lactic acid and energy**.
Used in making **bread and alcoholic drinks**.	Happens in muscles during **heavy exercise**.

Activity 5.5

- Take some fruit juice or sugar solution and add some yeast to this. Take this mixture in a test tube fitted with a one-holed cork.
- Fit the cork with a bent glass tube. Dip the free end of the glass tube into a test tube containing freshly prepared lime water.
- What change is observed in the lime water and how long does it take for this change to occur?
- What does this tell us about the products of fermentation?



PK Points:

- Fruit juice or sugar solution is taken because yeast needs sugar as food.
- Yeast is added to the sugar solution.
- The test tube is fitted with a cork and bent glass tube to pass the gas produced into lime water.
- The free end of the glass tube is dipped in fresh lime water to test the gas produced.
- Yeast breaks down sugar in the absence of oxygen.
- This process is called **fermentation**.
- During fermentation, carbon dioxide gas is released.
- The released CO_2 passes into lime water.
- Lime water turns milky due to carbon dioxide.
- This shows that **carbon dioxide is one of the products of fermentation.**

Q. Why do we get muscle cramps during intense exercise?

Ans. During intense exercise, our muscles need energy quickly. Sometimes, oxygen supply becomes insufficient, so muscles break down glucose without enough oxygen and form lactic acid.

The buildup of lactic acid in muscles causes fatigue and cramps.

LETS'S TEST YOUR MEMORY

Q. During vigorous exercise, muscles switch to anaerobic respiration even though oxygen is present in the body. Why?

A. Oxygen is completely absent

B. Oxygen supply is insufficient for demand

C. Muscles stop aerobic respiration permanently

D. Glucose is absent

LETS'S TEST YOUR MEMORY

Q. Which option correctly compares aerobic and anaerobic respiration?

Feature	Aerobic	Anaerobic
Energy yield	?	?

A. High / Low

B. Low / High

C. Equal / Equal

D. Zero / High

LETS'S TEST YOUR MEMORY

Q. The breakdown of glucose has taken the following pathway: (1 Mark) The sites 'a' and 'b' respectively are:

$$\text{Glucose} \xrightarrow{(a)} \text{Pyruvate} + \text{Energy} \xrightarrow{(b)} \text{Lactic acid} + \text{Energy}$$

A. Mitochondria and Oxygen deficient muscle cells

B. Cytoplasm and Yeast cells

C. Cytoplasm and Oxygen rich muscle cells

D. Cytoplasm and Oxygen deficient muscle cells

LETS'S TEST YOUR MEMORY

Q. In aerobic respiration, the steps are: breakdown of glucose to pyruvate and its further conversion to carbon dioxide. Both processes respectively occur in:

A. Vacuole and Cytoplasm

B. Chloroplast and Mitochondria

C. Mitochondria and Cytoplasm

D. Cytoplasm and Mitochondria

LETS'S TEST YOUR MEMORY

Q. Which situation best proves that breathing and respiration are independent processes?

A. Faster breathing during exercise

B. Holding breath for a few seconds

C. Slow breathing during sleep

D. Normal breathing but low energy levels

LETS'S TEST YOUR MEMORY

Q. A patient has normal breathing movements and oxygen intake, but cells show reduced energy release.

1. Identify the affected life process.
2. Why is breathing not sufficient here?
3. What would be the visible symptom?

1. Affected life process: Cellular respiration.
2. Reason: Breathing only brings oxygen into the body, but energy is released when food is broken down inside the cells. If cellular respiration is affected, energy release will reduce even if breathing is normal.
3. Visible symptom: Weakness, tiredness or fatigue.

LETS'S TEST YOUR MEMORY

Q. Which row in the table below shows the correct products of anaerobic respiration in humans and in yeast?

Row	Humans (Lactic Acid)	Humans (Carbon Dioxide)	Yeast (Lactic Acid)	Yeast (Carbon Dioxide)
1	✗	✓	✗	✗
2	✓	✗	✗	✓
3	✗	✓	✓	✗
4	✓	✓	✓	✗

LETS'S TEST YOUR MEMORY

Q. Sometimes sports man while running , the atletes suffer from muscle cramps why ? how is the respiration in this case is different from aerobic respiration .

Answer:

When sportspersons run or perform vigorous physical activities, the oxygen supply to their muscle cells becomes insufficient. As a result, the muscle cells perform anaerobic respiration, in which glucose is partially broken down into lactic acid. The accumulation of lactic acid in the muscles causes muscle cramps.

This type of respiration is different from aerobic respiration because it occurs without oxygen, produces less energy, and results in lactic acid as a byproduct. In contrast, aerobic respiration takes place in the presence of oxygen, produces more energy, and forms carbon dioxide and water.

LETS’S TEST YOUR MEMORY

NCERT

Q. What are the differences between aerobic and anaerobic respiration? Name some organisms that use the anaerobic mode of respiration..

Aerobic Respiration	Anaerobic Respiration
It takes place in the **presence of oxygen**.	It takes place in the **absence of oxygen**.
Glucose is **completely broken down**.	Glucose is **incompletely broken down**.
End products are **carbon dioxide, water and energy**.	End products may be **ethanol, carbon dioxide and energy** or **lactic acid and energy**.
It releases **more energy**.	It releases **less energy**.
It mainly occurs in **mitochondria**.	It occurs in **cytoplasm**.

Yeast uses anaerobic respiration during fermentation.
Some bacteria also use anaerobic respiration.

LETS’S TEST YOUR MEMORY

Q. Ravi is a 16-year-old athlete preparing for his school sports competition. During practice, he notices that his breathing rate increases significantly while running, and sometimes he feels mild cramps in his legs. His coach explains that the body requires more energy during intense activity, and oxygen supply may not meet the demand.

<u>Questions:</u>

- Why does Ravi’s breathing rate increase during exercise?
- Name the type of respiration that occurs in muscles during intense physical activity.
- What causes cramps in muscles after heavy exercise?
- Write the word equation for anaerobic respiration in human muscles.
- Suggest one way to reduce muscle cramps.

LETS'S TEST YOUR MEMORY

Q. All living cells require energy for various activities. This energy is available by the breakdown of simple carbohydrates either using oxygen or without using oxygen.

The graph below represents the blood lactic acid concentration of an athlete during a race of 400 m and shows a peak at point D.

Lactic acid production has occurred in the athlete while running in the 400 m race.

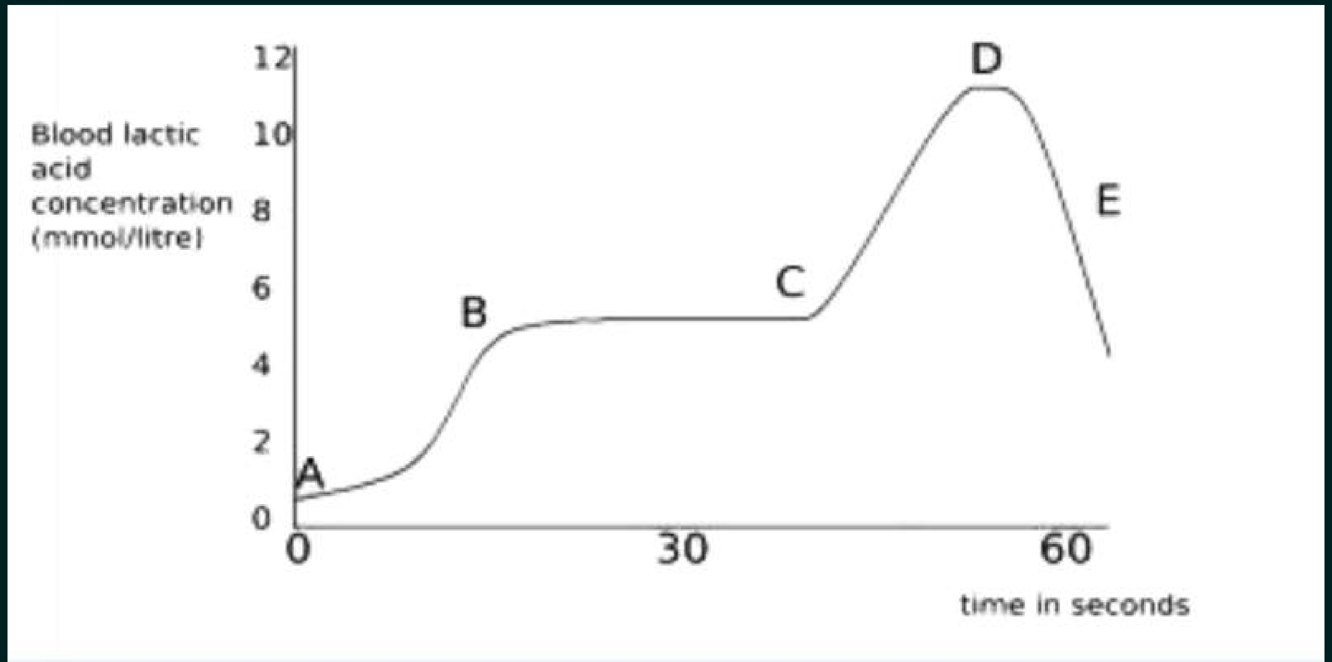
(i) Which of the following processes explains this event?

A. Aerobic respiration

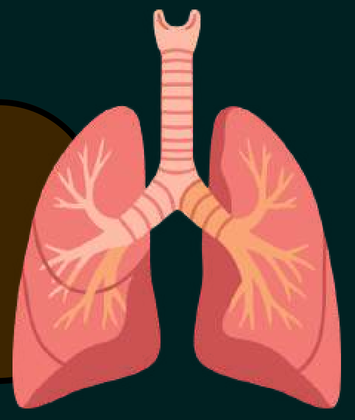
B. Anaerobic respiration

C. Fermentation

D. Breathing



Breathing vs Respiration

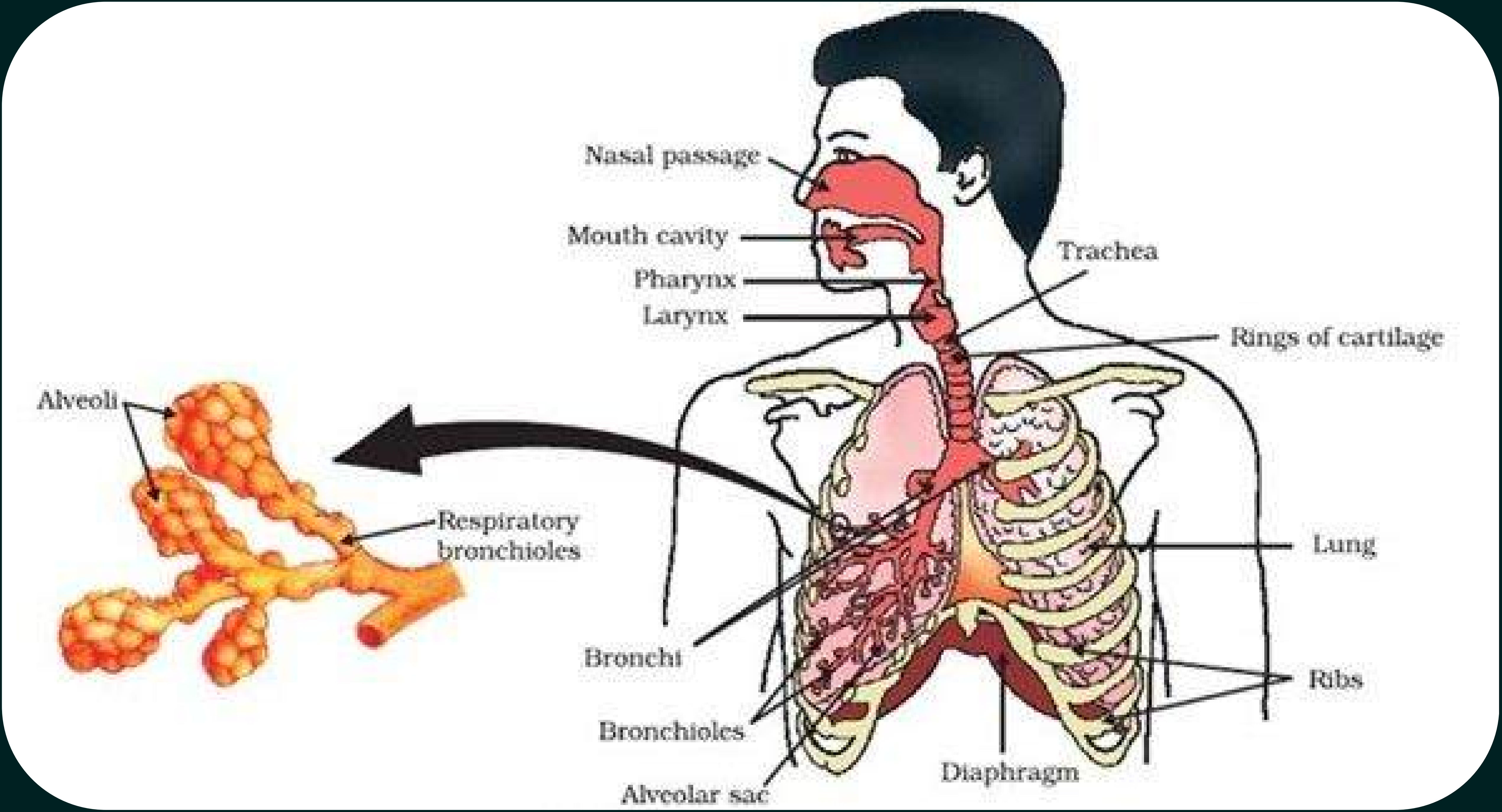


Breathing	Respiration
Breathing is the process by which oxygen is taken into the body and carbon dioxide is released out.	Respiration is the process in which food is broken down inside the cells to release energy.
It is a physical process.	It is a biochemical process.
It occurs in respiratory organs like lungs.	It occurs inside the cells.
It involves exchange of gases.	It involves breakdown of glucose.
Energy is not released during breathing.	Energy is released during respiration.
Example: Inhalation and exhalation.	Example: Breakdown of glucose to release energy.

आरंभ BATCH

Human Respiratory System

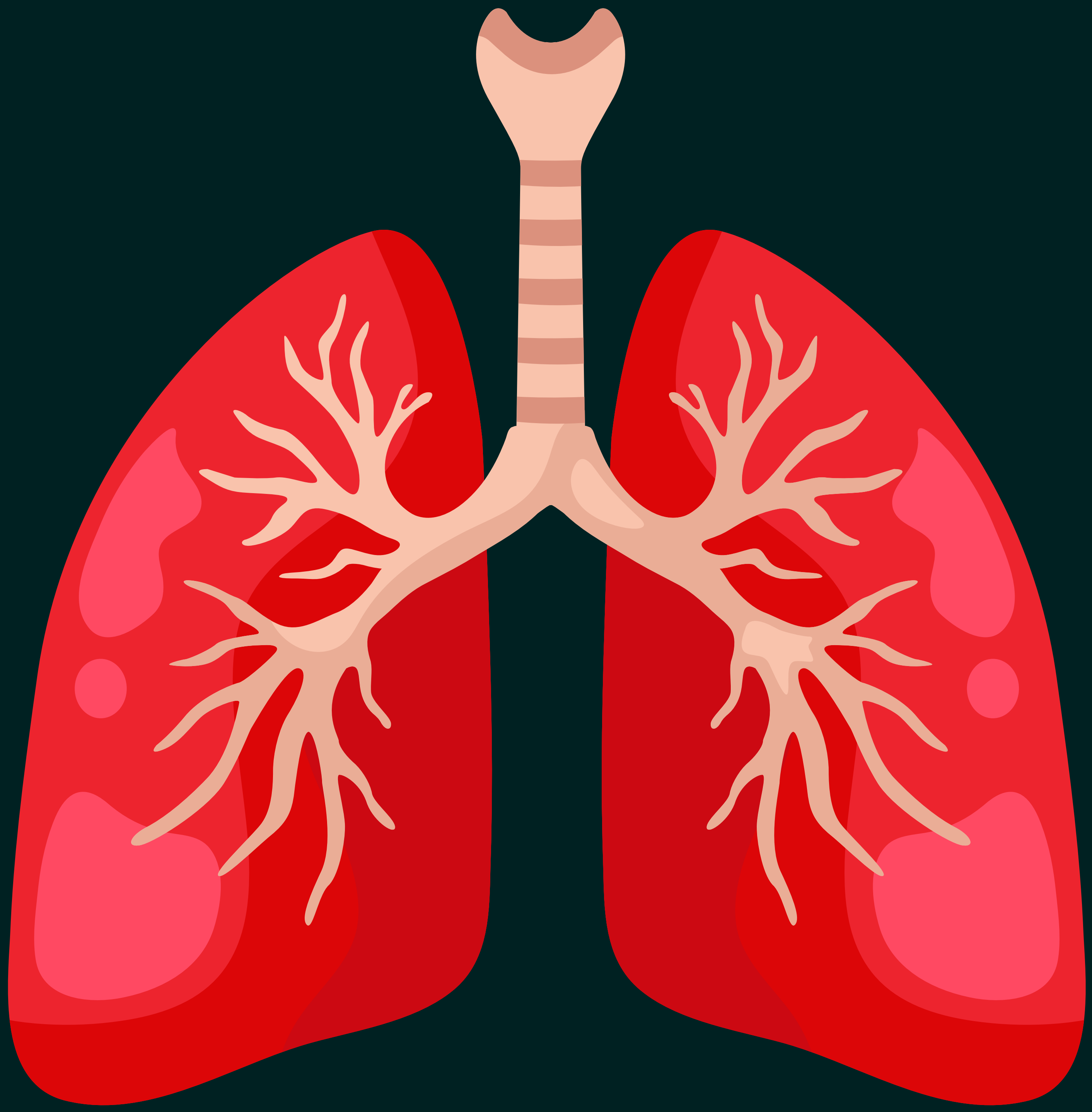
Human Respiratory System



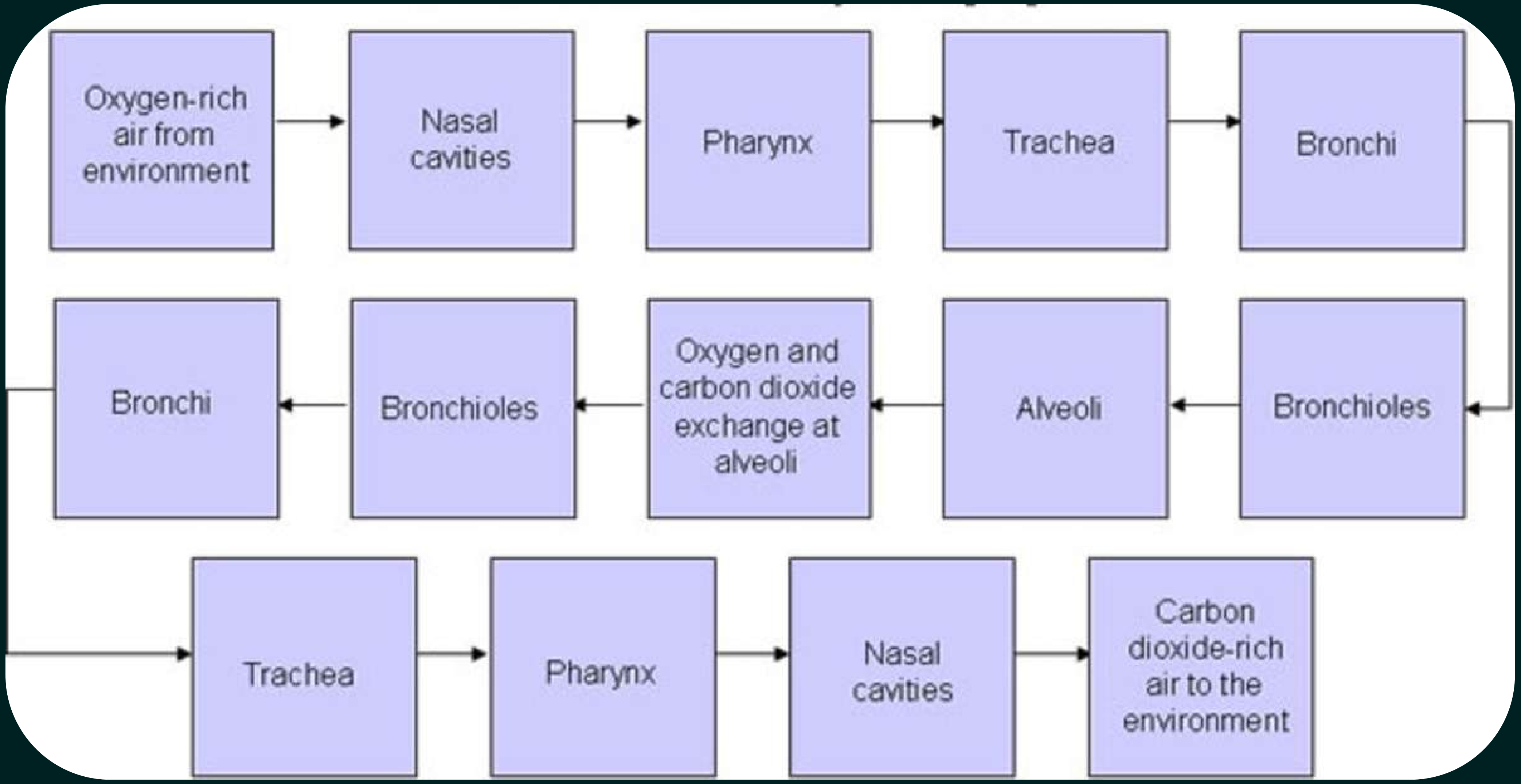
Human Respiratory System

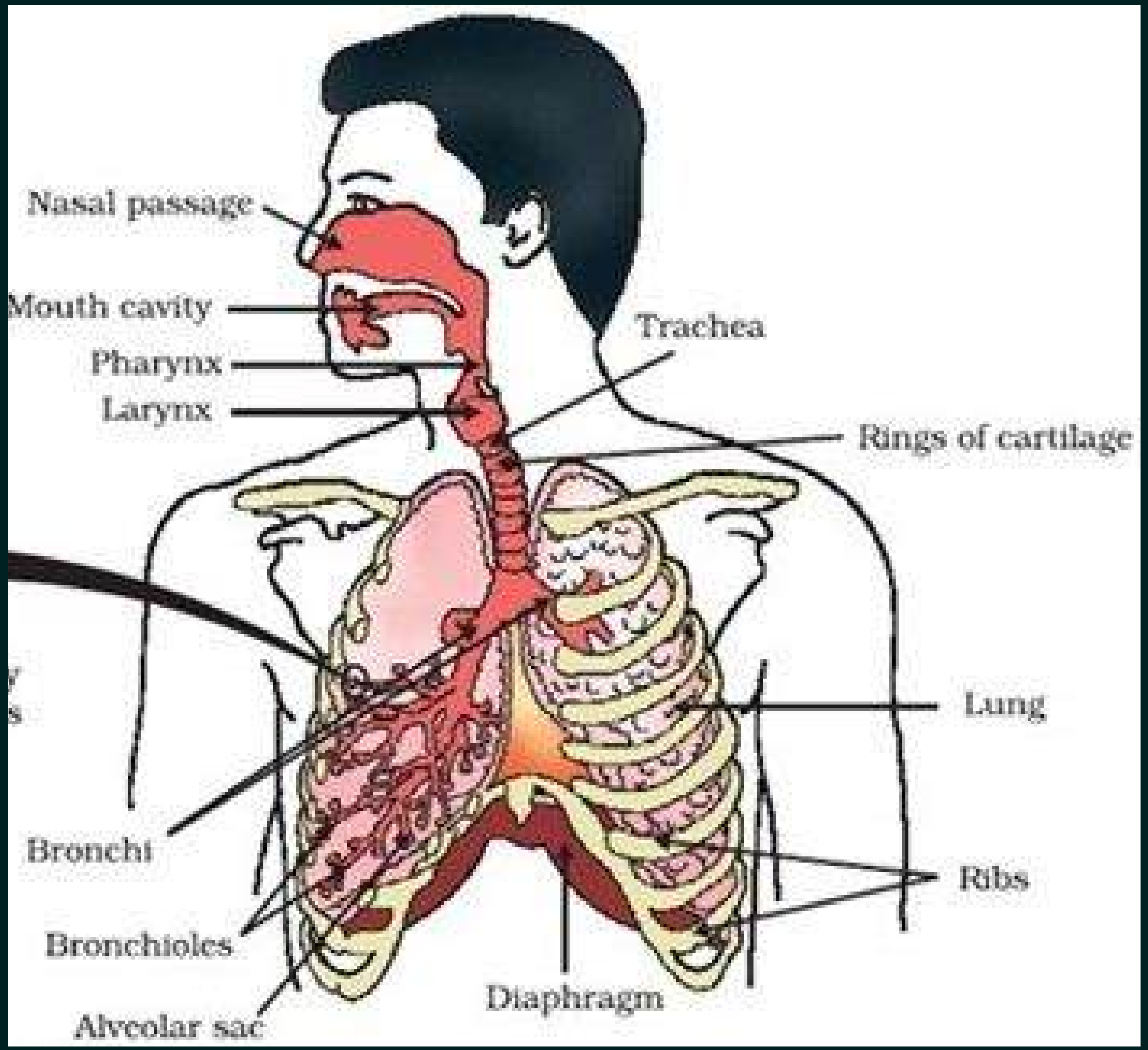
- The human respiratory system is the biological system responsible for taking in oxygen and removing carbon dioxide from the body.

- It helps us breathe, and it's essential for cellular respiration, which produces the energy our body needs to function.



Exchange of gases in Human Body





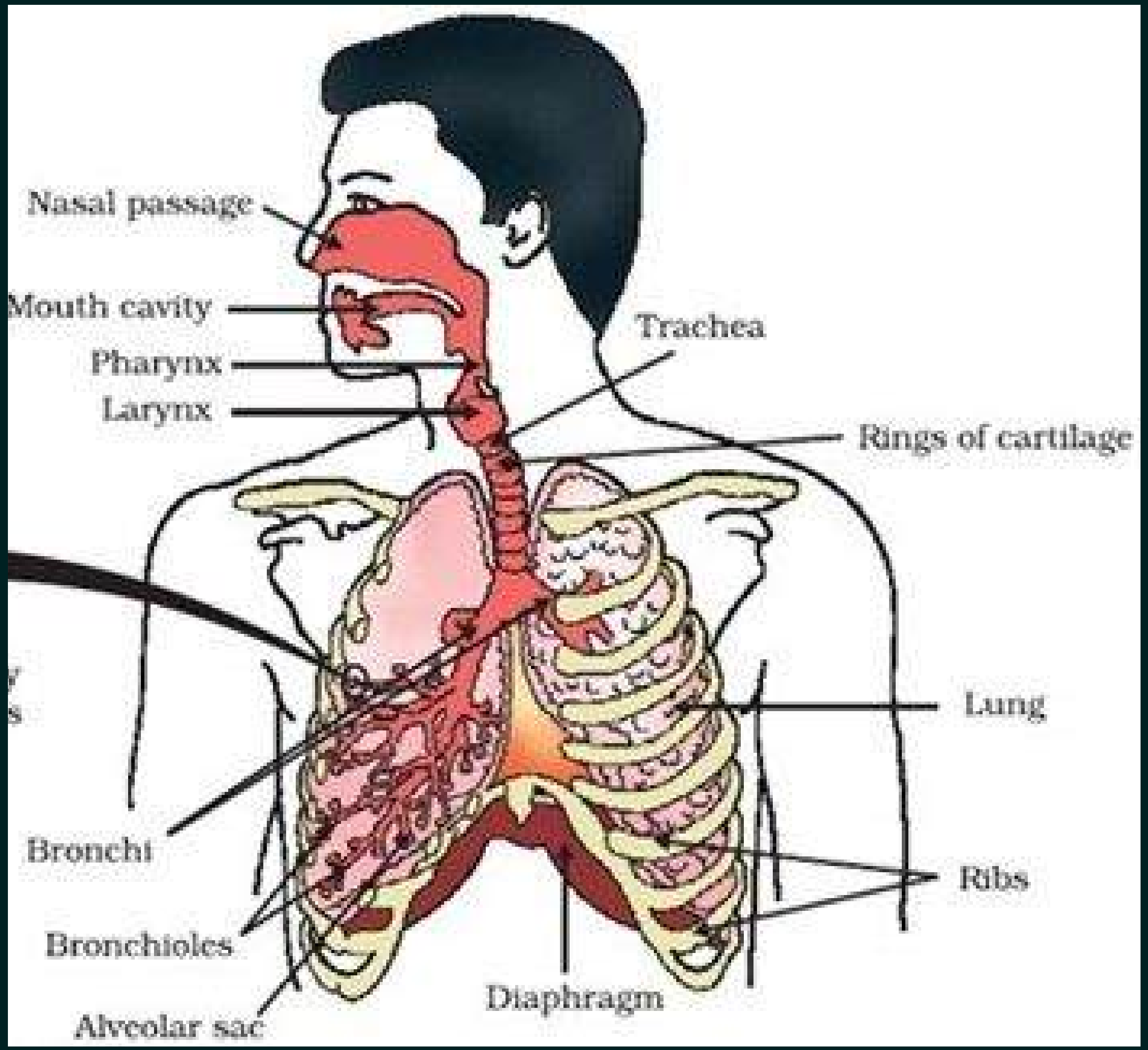
Passage of air through Respiratory System:

Nostril: Air enters the body through the nostrils.

Nasal Passage: The air passing through the nostrils is filtered by fine hairs that line the passage.
The passage is also lined with mucus, which helps in trapping dust and other particles.

Pharynx: Pharynx is the common passage for food and air. Air from the nasal passage passes through the pharynx and moves towards the windpipe.

Larynx: Larynx is also called the voice box.
It connects the pharynx to the trachea and helps in producing sound.

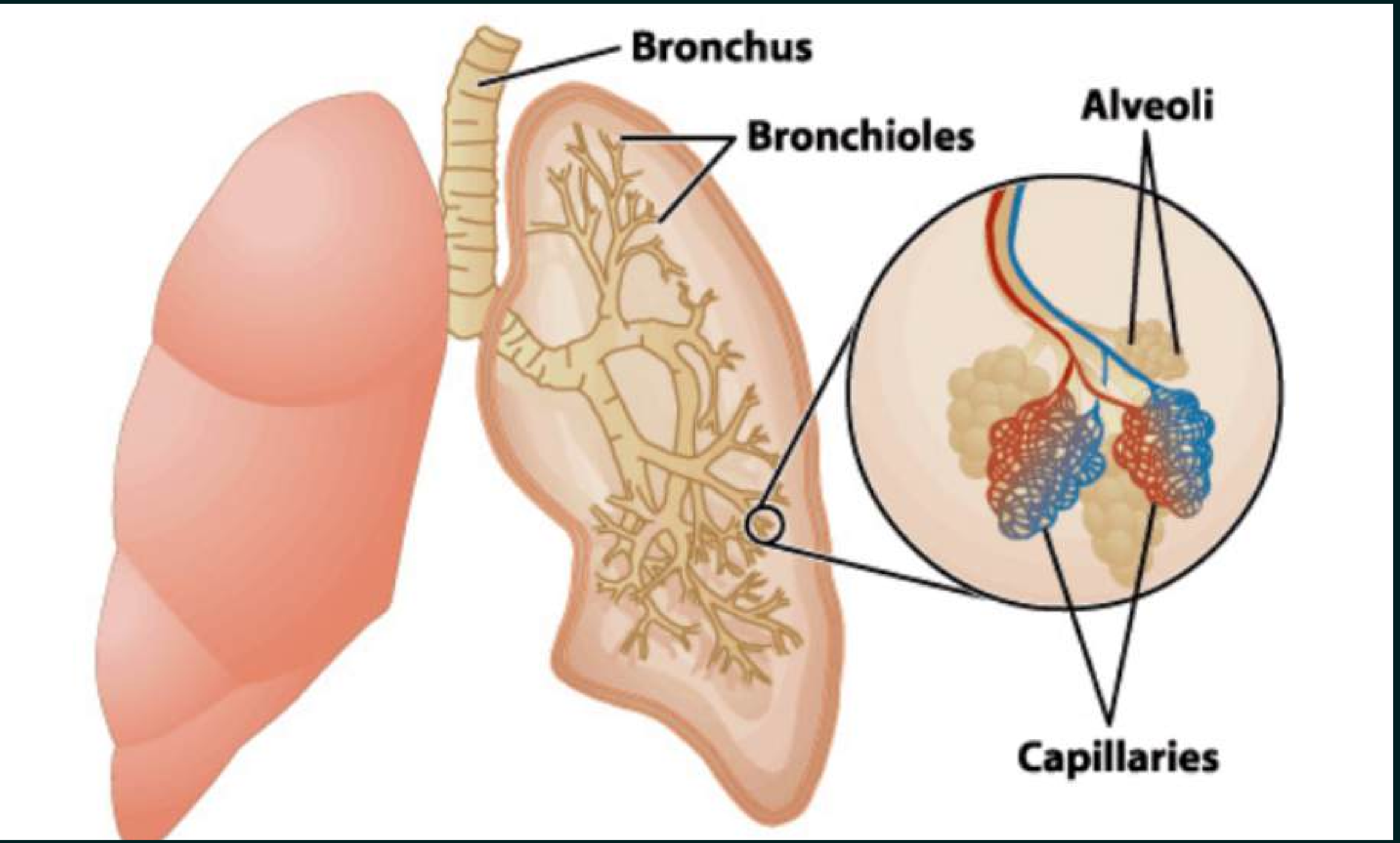


Trachea / Air Passage: Trachea is the windpipe through which air moves towards the lungs.
Rings of cartilage are present in the trachea/air passage.
These rings prevent the air passage from collapsing.

Bronchi: The trachea divides into two bronchi, one entering each lung.

Bronchioles: Inside the lungs, the bronchi divide into smaller and smaller tubes called bronchioles

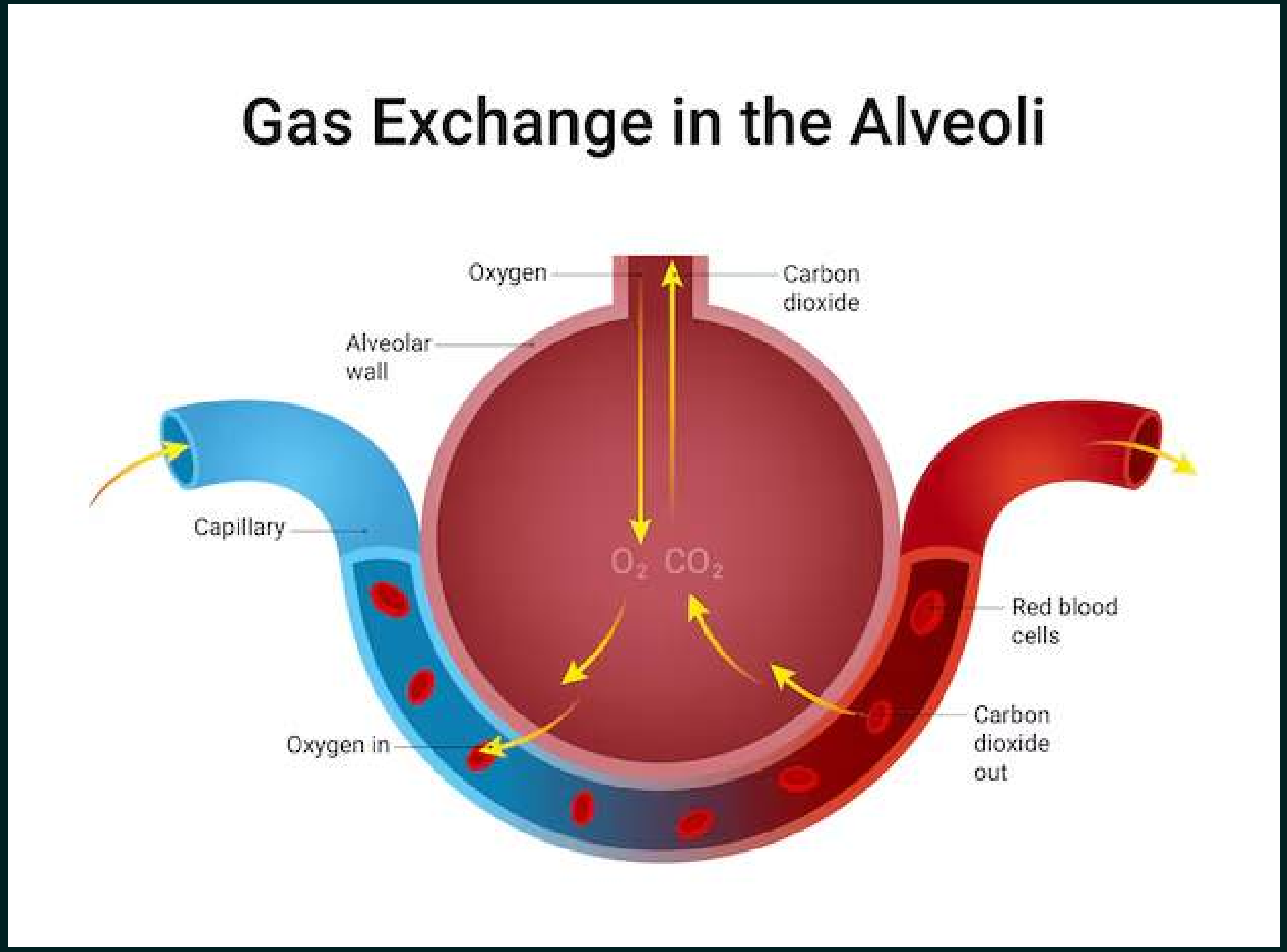
Alveoli: Bronchioles finally end in balloon-like structures called alveoli.
Alveoli provide a surface where exchange of gases takes place.



They are the site of gas exchange in the lungs

Alveolus Gas Exchange

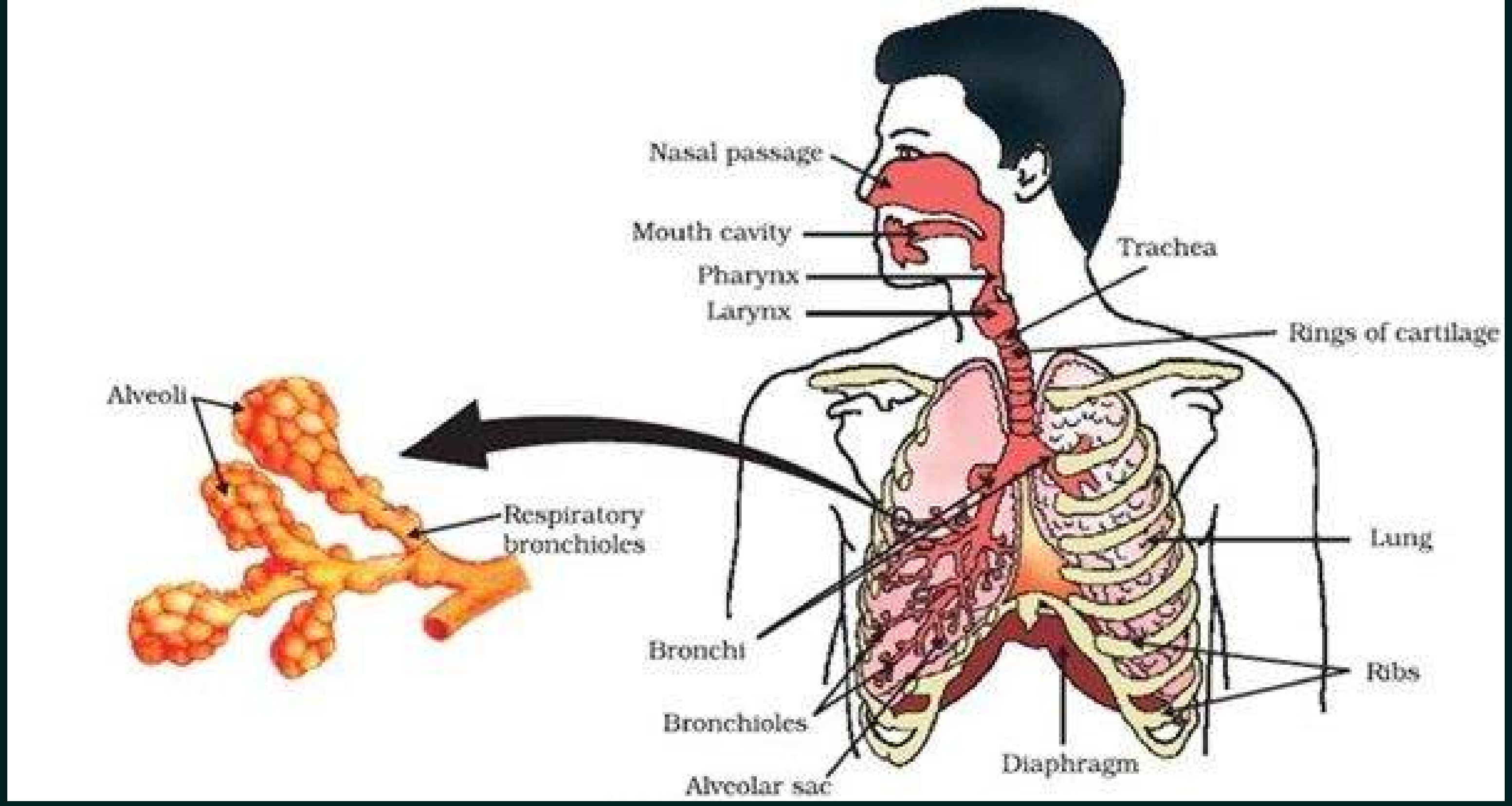
The blood brings carbon dioxide from the rest of the body for release into the alveoli, and the oxygen in the alveolar air is taken up by blood in the vessels to be transported to all the cells in the body.



Alveoli are suitable for gas exchange because:

1. **They provide a large surface area.**
2. **They have thin walls.**
3. **They are surrounded by blood capillaries.**
4. **They have moist surfaces for diffusion of gases.**

Blood Vessels Around Alveoli: The walls of alveoli contain an extensive network of blood vessels.
Oxygen from the alveolar air is taken up by blood.
Carbon dioxide from the blood is released into the alveoli.



Haemoglobin

- A respiratory pigment
- Iron containing protein
- Gives red color to red blood cells.

Affinity of Haemoglobin for various gases: $CO > O_2 > CO_2$

Red Blood Cells

- Haemoglobin present inside RBCs binds with oxygen.
- It takes up oxygen from the lungs.
- RBCs transport oxygen to all parts of the body.
- After reaching body tissues, haemoglobin releases oxygen for cellular respiration.
- Carbon dioxide is mostly transported in dissolved form in the blood because it is more soluble in water than oxygen.

Fact: The biconcave shape of RBCs increases surface area for efficient gas exchange!

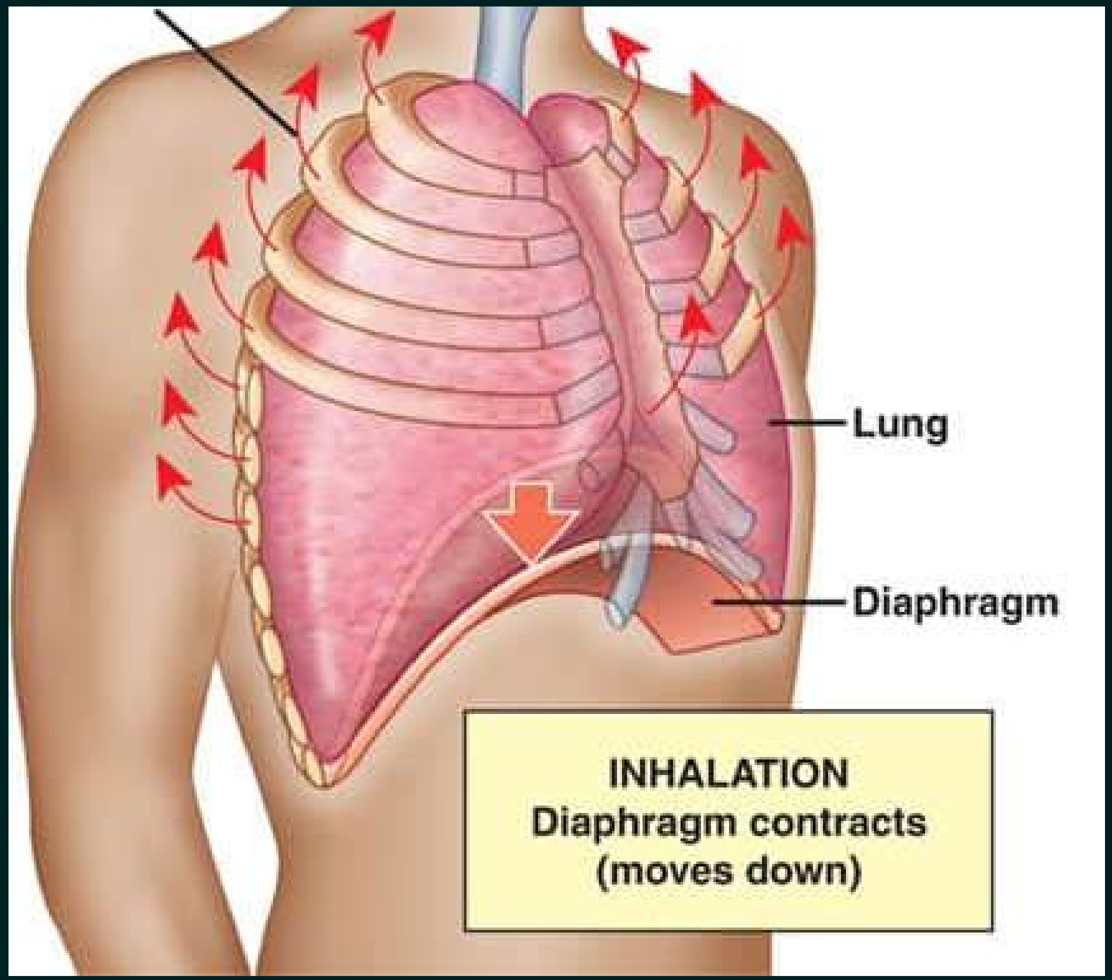
Soch ke bataao?

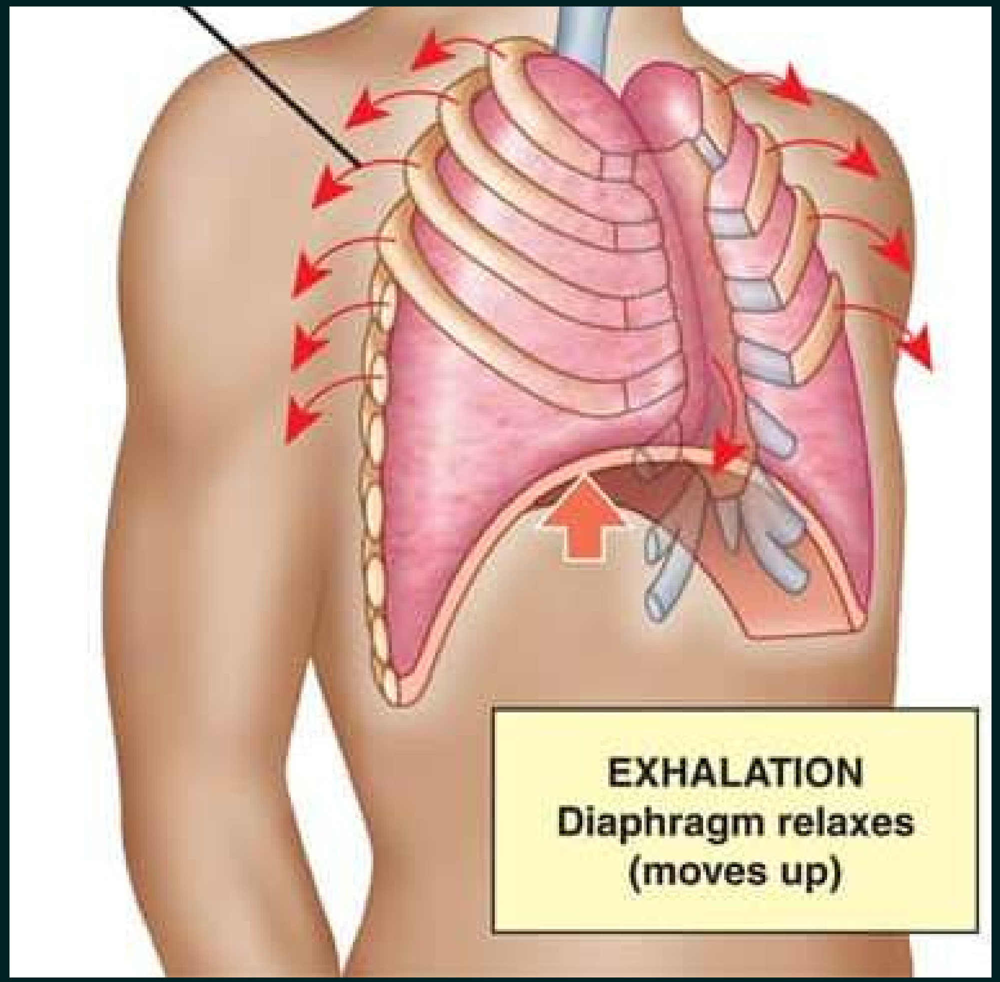
Lack of hemoglobin will lead to weakness?

YES or NO??

Breathing Mechanism (Inhalation and Exhalation)

DIAPHRAGM: A skeletal muscle that separates the chest from the abdomen and helps in breathing.





Inhalation: Ribs move up, diaphragm flattens
Chest cavity increases → air is sucked into lungs

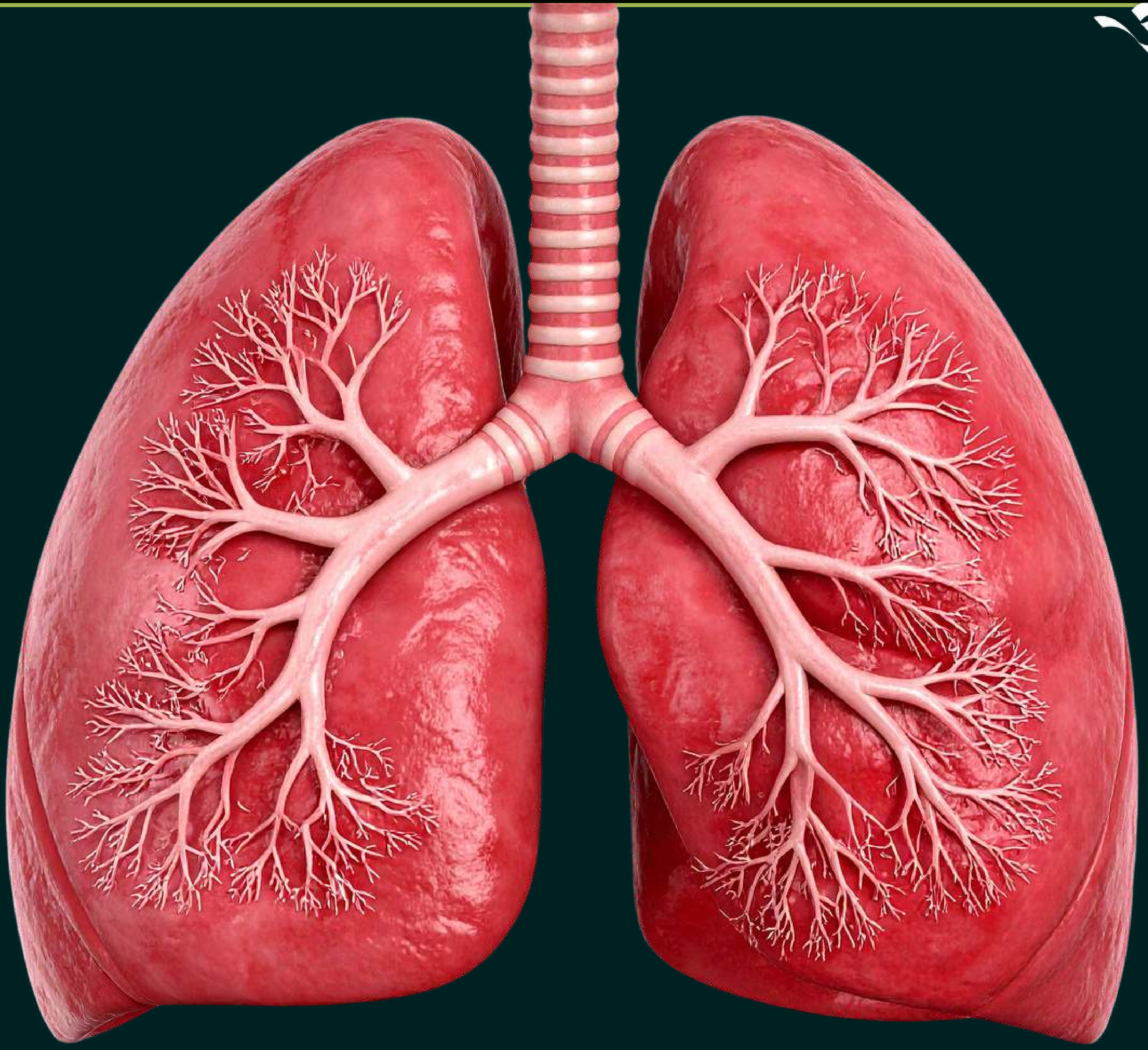
Exhalation: Ribs move down, diaphragm relaxes (domes up)
Chest cavity decreases → air is pushed out

Inhalation V/S Exhalation

Inhalation	Exhalation
It is the process of breathing in air.	It is the process of breathing out air.
Ribs move upwards and outwards.	Ribs move downwards and inwards.
Diaphragm contracts and flattens.	Diaphragm relaxes and becomes dome-shaped/arches upwards.
Volume of the chest cavity increases.	Volume of the chest cavity decreases.
Air pressure inside the chest cavity decreases.	Air pressure inside the chest cavity increases.
Air is sucked into the lungs.	Air is pushed out of the lungs.

Q. Why lungs have residual air

The lungs always contain some residual air so that there is enough time for oxygen to be absorbed into the blood and carbon dioxide to be released from the blood into the alveoli.



LETS'S TEST YOUR MEMORY

Q. One-cell thick blood vessels are known as:

A. Alveoli

B. Capillaries

C. Arteries

D. Veins

LETS'S TEST YOUR MEMORY

Q. A person has normal lung volume but reduced oxygen absorption. Which is the most likely cause?

A. Increased breathing rate

B. Reduced alveolar surface area

C. Increased diaphragm movement

D. Increased chest expansion

LETS'S TEST YOUR MEMORY

Q. During heavy exercise, breathing becomes rapid. However, oxygen delivery still becomes limiting. Why?

A. Increasing Lungs stop working gradient

B. Blood cannot carry oxygen fast enough

C. Oxygen is not present

D. Alveoli collapse

LETS'S TEST YOUR MEMORY

Q. If alveolar walls become thicker, which immediate change occurs?

A. Increased diffusion rate

B. Decreased diffusion rate

C. No effect

D. Increased lung capacity

LETS'S TEST YOUR MEMORY

Q. Which of the following statements are correct in reference to the role of A, shown in the given diagram, during a breathing cycle in human beings?

(i) It helps to decrease the residual volume of air in lungs.

(ii) It flattens as we inhale.

(iii) It gets raised as we inhale.

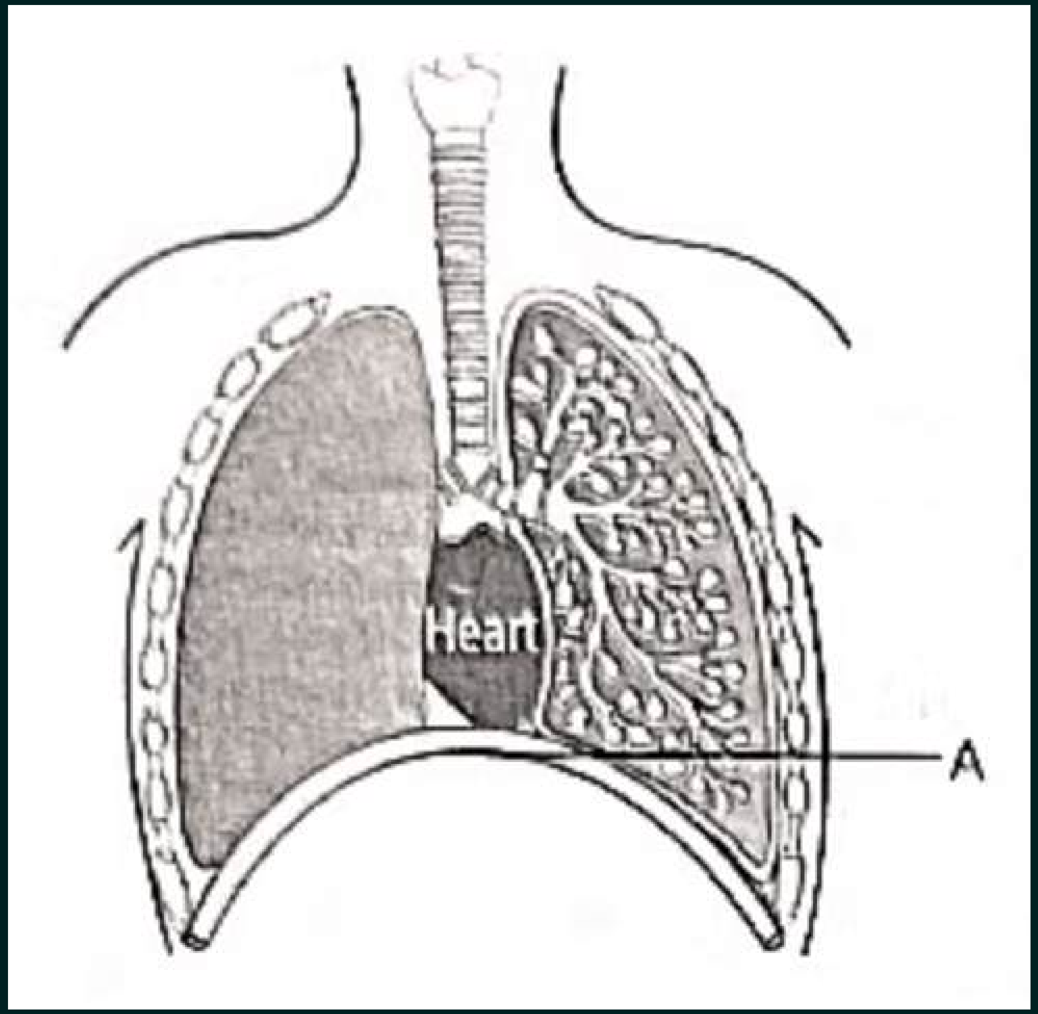
(iv) It helps the chest cavity to become larger.

(a) (ii) and (iv)

(b) (iii) and (iv)

(c) (i) and (iii)

(d) (i), (ii) and (iv)



LETS'S TEST YOUR MEMORY

Q. Two individuals inhale the same volume of air:

- Person A: Healthy lungs
- Person B: Reduced alveolar surface

Q1. Who absorbs more oxygen? Why?

Q2. Why can't Person B compensate by breathing faster?

1. Because healthy lungs have a larger alveolar surface area, so more oxygen diffuses into the blood.
2. because the alveolar surface area is reduced. So, even if more air enters, less oxygen can pass into the blood.

LETS'S TEST YOUR MEMORY

Q. Why is surface area more important than breathing rate for gas exchange?

Surface area is more important because the exchange of gases takes place through the surface of alveoli.

If alveolar surface area is large, more oxygen can diffuse into the blood at the same time. But if the surface area is reduced, then even faster breathing cannot help much because less area is available for gas exchange.

LETS'S TEST YOUR MEMORY

Q. Why does breathing continue even when oxygen demand is low (like during sleep)?

Breathing continues even during sleep because body cells still need oxygen for cellular respiration and continue producing carbon dioxide.
Even when oxygen demand is low, oxygen is needed to release energy from food, and carbon dioxide must be removed from the body.

LETS'S TEST YOUR MEMORY

Q. Predict what happens if alveoli lose moisture.

If alveoli lose moisture, the diffusion of gases will become less efficient. Oxygen will not dissolve properly and pass into the blood easily. Similarly, carbon dioxide removal will also become slow.

LETS'S TEST YOUR MEMORY

Q. Give reasons for the following:

(a) Rings of cartilage are present in the throat.

(b) Lungs always contain a residual volume of air.

(c) The diaphragm flattens and ribs are lifted up when we breathe in.

(d) Walls of alveoli contain an extensive network of blood vessels.

Ans. (a) To keep the windpipe open and prevent its collapse, ensuring continuous airflow to the lungs.

(b) To allow continuous exchange of gases and prevent lung collapse.

(c) To increase chest cavity volume, creating low pressure so air can enter the lungs.

(d) To allow efficient exchange of oxygen and carbon dioxide between lungs and blood.

LETS'S TEST YOUR MEMORY

Q. Why is respiratory pigment needed in multicellular organisms with large body sizes?

Ans. In multicellular organisms with large body sizes, diffusion alone is not sufficient to supply oxygen to all body cells efficiently. Hence, a respiratory pigment like haemoglobin is required.
Haemoglobin, present in red blood cells (RBCs), has a high affinity for oxygen.
It helps in absorbing oxygen from the lungs and transporting it to all body tissues.
It also carries carbon dioxide from tissues back to the lungs for exhalation

Exchange of gases in Plants

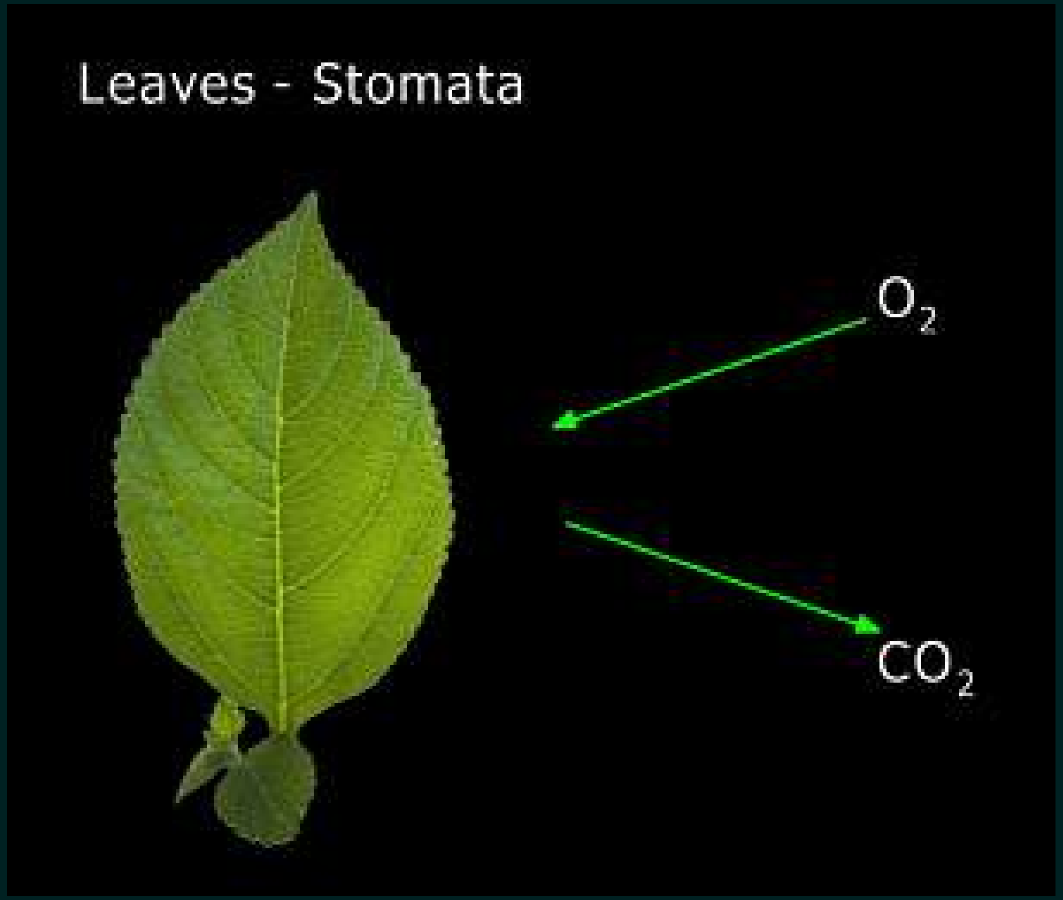
Respiration in plants is simpler than the respiration in animals.

Gaseous exchange occur through :

(a) Stomata in leaves

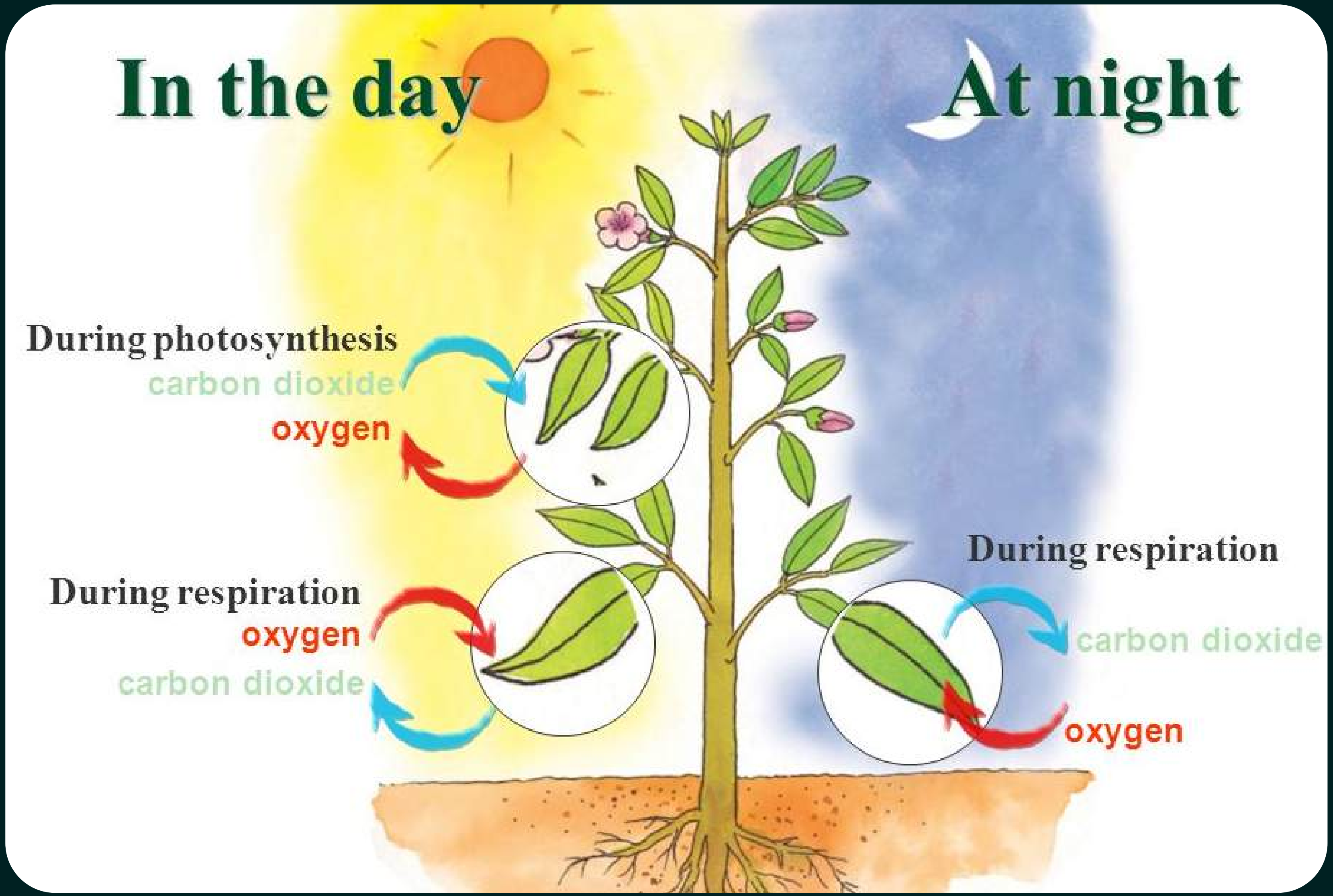
(b) Lenticles in stems

(c) General surface of Roots



Exchange of gases in Plants

Daytime 🌞	Night time 🌙
Photosynthesis + Respiration	Only Respiration
CO_2 is taken in for photosynthesis	CO_2 is released as a byproduct of respiration
O_2 is released as a byproduct of photosynthesis	O_2 is used for respiration
Oxygen released, CO_2 absorbed	Carbon dioxide released, O_2 absorbed



Roots take oxygen from the air spaces present between soil particles. Carbon dioxide produced during respiration diffuses out through the roots.

LETS'S TEST YOUR MEMORY

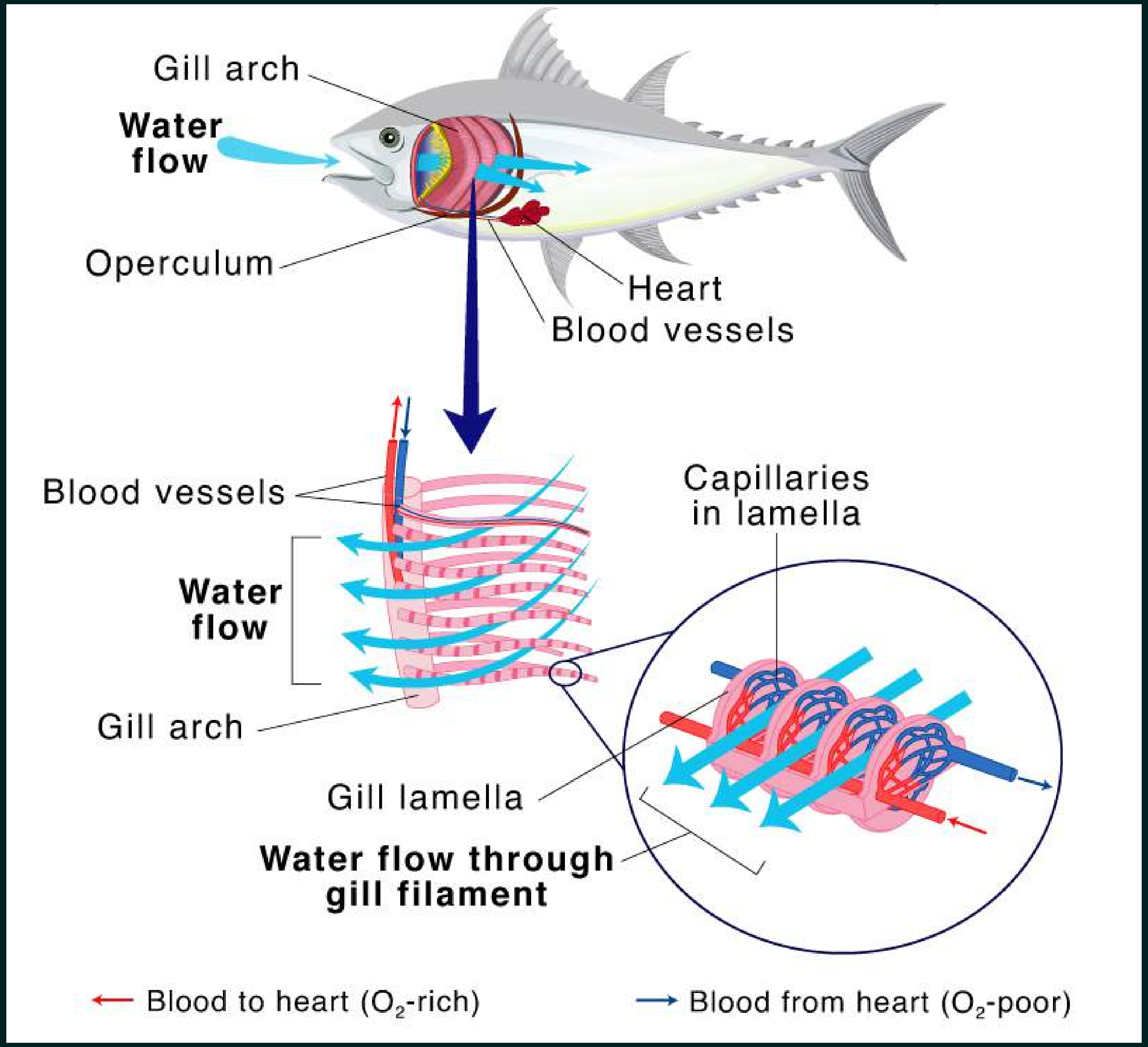
Q. Why is waterlogging harmful for plants?

Answer:

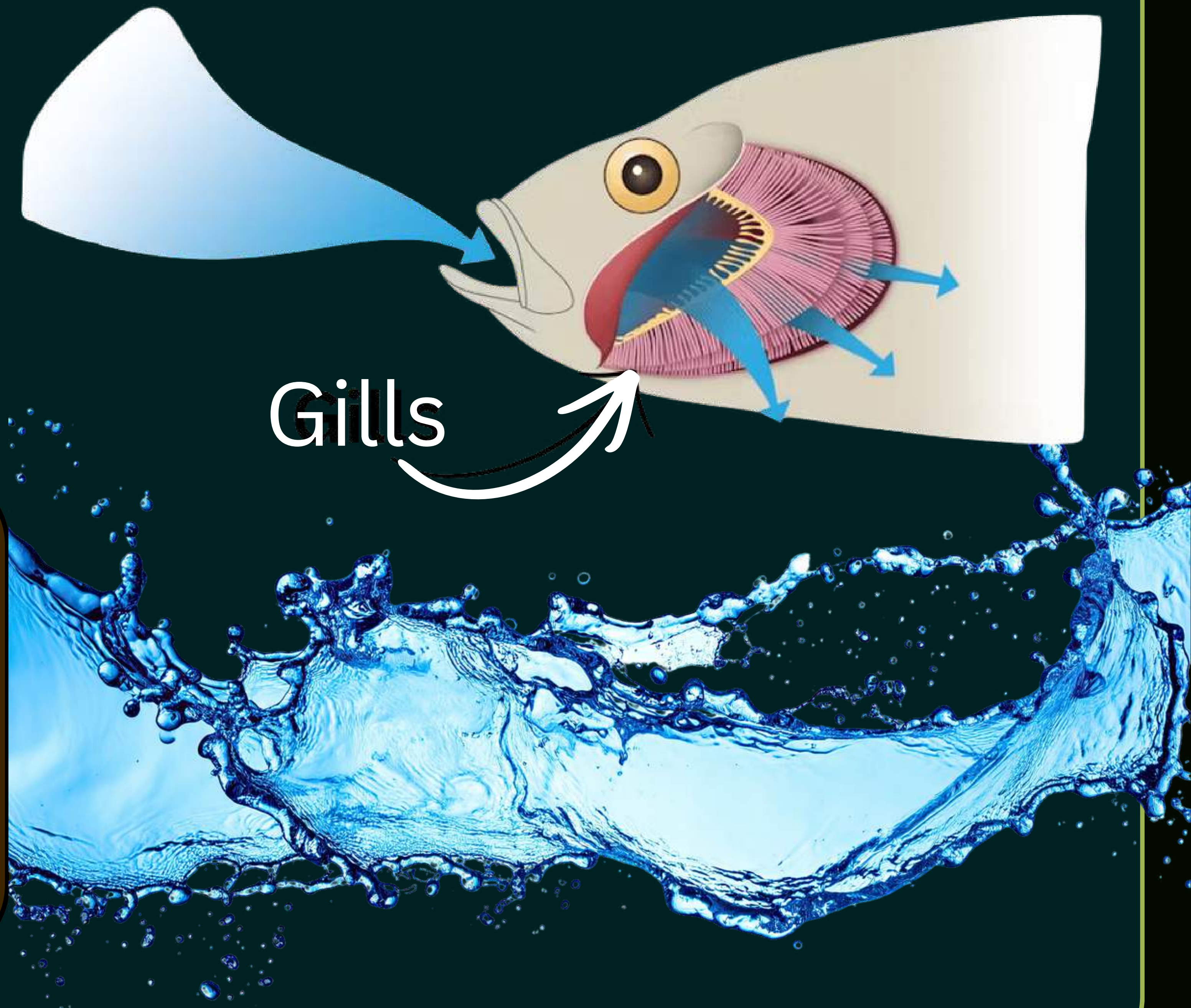
Waterlogging is harmful for plants because water fills the air spaces present between soil particles. Due to this, roots do not get enough oxygen for respiration. Without proper respiration, roots cannot release enough energy, so the plant may stop growing and can even die.

How do fishes breathe underwater?

- Fishes use **gills** as their respiratory organs.
- Gills are located on the sides of the head and are richly supplied with **blood capillaries.**
- When a fish opens its mouth, water enters and passes over the gills.
- The gills absorb oxygen dissolved in water and transfer it into the blood through diffusion in the capillaries. At the same time, carbon dioxide from the blood diffuses out into the water and is removed.
- The gills are covered by a bony flap called the **operculum**, which protects the delicate gill filaments and also helps maintain a steady flow of water over them, even when the fish is not swimming.



- In fishes, the blood passes only through the gills, not through the rest of the body during the process of oxygenation. This is different from double circulation seen in humans. The oxygenated blood is then circulated to the rest of the body.
- This system allows fishes to breathe and survive underwater, even though there is no free oxygen gas like in the air.



Since the amount of dissolved oxygen is fairly low compared to the amount of oxygen in the air, the rate of breathing in aquatic organisms is much faster than that seen in terrestrial organisms

LETS'S TEST YOUR MEMORY

Q. As compared to terrestrial organisms, the rate of breathing in aquatic organisms is:

A. faster because they need more oxygen for their survival

B. faster because the amount of dissolved oxygen in water is fairly low

C. slower because the amount of dissolved oxygen in water is fairly low

D. slower because the capacity of water of dissolving atmospheric air is limited

LETS'S TEST YOUR MEMORY

Q. What advantage over an aquatic organism does a terrestrial organism have with regard to obtaining oxygen for respiration? [NCERT]

Answer:

Oxygen is more abundant in the air than in water.

Terrestrial organisms get oxygen more easily due to better respiratory surfaces (lungs, skin, etc.).

LETS'S TEST YOUR MEMORY

Q. Rate of breathing in aquatic organisms is much faster than that in terrestrial organisms. Give reasons.

The rate of breathing in aquatic organisms is much faster because the amount of dissolved oxygen in water is much less than the amount of oxygen present in air.
So, aquatic organisms have to breathe faster to take in enough oxygen for respiration.

LETS'S TEST YOUR MEMORY

Q. Aerobic respiration requires intake of oxygen to break down food and release energy. In plants and human beings, exchange of gases takes place through special structures. In humans, breathing helps air reach the lungs, where exchange of gases takes place. The diaphragm helps in changing the size of the chest cavity. In plants, exchange of gases mainly occurs by diffusion. However, in large animals like humans, diffusion alone is not sufficient to transport oxygen to all parts of the body. Therefore, oxygen is transported with the help of blood and haemoglobin.

Q1. Name the structures through which gaseous exchange takes place in:

(a) Plants

(b) Human beings

Q2. Name the structure that controls the size of the chest cavity in humans.

Q3. What is the process by which gas exchange occurs in plants?

Q4. Why is diffusion alone not sufficient to carry oxygen throughout the human body?

Q5. How is oxygen transported to all parts of the body in humans?

IT IS OKAY TO

- ASK QUESTIONS
- MAKE MISTAKES
- NOT KNOW IT ALL
- ASK FOR HELP
- START OVER AGAIN
- BE YOURSELF
- HAVE A BAD DAY